

Outline of the lecture

- MINIMIZATION TECHNIQUE: K-MAP
- EXPANSION INTO CANONICAL FORM
 - SOP (SUM OF PRODUCT)
 - POS (PRODUCT OF SUM)
- KARNAUGH MAPS (K-MAP)
 - MINIMIZATION OF SOP EXPRESSION
 - MINIMIZATION OF POS EXPRESSION
 - TWO VARIABLE K-MAP
 - THREE VARIABLE K-MAP
 - FOUR VARIABLE K-MAP
 - FIVE VARIABLE K-MAP
 - DON'T CARE CONDITION
- PROBLEMS FOR PRACTICE

MINIMIZATION TECHNIQUES

- ▶ Every logic operation represents an element of hardware. Reducing the expression to minimal, results in reduction of cost and complexity
- ▶ Boolean algebra uses laws and theorems to reduce but
 - ▶ it can differ from programmer to programmer
 - ▶ It is not sure that the minimal expression is really minimal
- ▶ Karnaugh map (K-map) is a chart or graph, composed of an arrangement of adjacent cells, each representing a particular combination of all the inputs.
 - ▶ For a 'n' variable input function, the number of cells in a K-map will be 2^n , i.e. for 2-variable, there will be 4 cells ; for 3-variable, there will be 8 cells in the K-map and so on.
 - ▶ Provide solution in most simplified form.

EXPANSION INTO CANONICAL FORM

- ▶ Any Boolean expression can be expressed in a Standard or Canonical or Expanded form.
- ▶ Difference between the Reduced Boolean expression and Canonical Boolean expression is that in the latter, each of the term comprises of all the input variables of the function.
- ▶ The expression can either be
 - ▶ Canonical Sum of Products (SOP)
 - ▶ Canonical Product of Sums (POS)

CANONICAL FORM-SOP

► Canonical Sum of Products (SOP):

- Number of product terms, each containing all the input variables in complemented or non-complemented form, are summed together.

E.g.: $f(A,B,C) = A'BC + AB'C' + ABC' + ABC$

- Each product term is called a MINTERM (m)
- To convert a minterm into binary word
 - The non-complemented variable is represented by 1
 - The complemented variable is represented by 0
- Minterm is represented as 'm' with decimal equivalent as subscript.
- E.g.: $f(A,B,C) = A'BC + AB'C' + ABC' + ABC = \sum (m_3, m_4, m_6, m_7)$
 $= \sum m(3,4,6,7)$

A'BC	AB'C'	ABC'	ABC
011=3	100=4	110=6	111=7
m ₃	m ₄	m ₆	m ₇

CANONICAL FORM-POS

► Canonical Product of Sums (POS):

- Number of sum terms, each containing all the input variables in complemented or non-complemented form, are multiplied together.

E.g.: $F(A,B,C) = (A'+B+C)(A+B'+C')(A+B+C')(A+B+C)$

- Each sum term is called a MAXTERM (M)
- To convert a maxterm into binary word
 - The non-complemented variable is represented by 0
 - The complemented variable is represented by 1
- Maxterm is represented as 'M' with decimal equivalent as subscript.
- E.g.: $F(A,B,C) = (A'+B+C)(A+B'+C')(A+B+C')(A+B+C)$
 $= \prod (M_0, M_1, M_3, M_4) = \prod M(0,1,3,4)$

$(A'+B+C)$	$(A+B'+C')$	$(A+B+C')$	$(A+B+C)$
100=4	011=3	001=1	000=0
M_4	M_3	M_1	M_0

EXPANSION INTO CANONICAL FORM

► The steps to expand an expression into Canonical Sum of Products (SOP):

1. Write all the terms
2. For each missing variable in the product term, multiply it with the sum of the missing variable and its complement. Now expand the expression.
3. Drop the redundant/ duplicate terms.

► E.g.: Expand into Canonical SOP, $f(A,B) = A' + B'$

$$\begin{aligned}A' + B' &= A'(B+B') + B'(A+A') \\&= A'B + A'B' + B'A + B'A' \\&= A'B + A'B' + AB' \\&= 01 + 00 + 10 \\&= m_1 + m_0 + m_2 \\&= \sum m(0,1,2) = \prod M(3)\end{aligned}$$

A	B	O/P
0	0	1
0	1	1
1	0	1
1	1	0

- The missing minterms of canonical SOP are the terms present in canonical POS.
- Minterms in the expression signify that the output of the circuit assumes a logic 1 for those minterms/ input combinations.

EXPANSION INTO CANONICAL FORM

► The steps to expand an expression into Canonical Product of Sums (POS):

1. Write all the terms
2. For each missing variable in the sum term, add it with the product of the missing variable and its complement. Now expand the expression.
3. Drop the redundant/ duplicate terms.

► E.g.: Expand into Canonical POS, $F(A,B) = A.(A + B').B$

$$\begin{aligned} A.(A + B').B &= (A + B.B')(A + B')(B + AA') \\ &= (A+B)(A+B')(A+B')(B+ A)(B+A') \\ &= (A+B)(A+B')(A'+B) \\ &= (00) (01) (10) \\ &= M_0 . M_1 . M_2 \\ &= \prod M(0,1,2) = \sum m(3) \end{aligned}$$

A	B	O/P
0	0	0
0	1	0
1	0	0
1	1	1

- The missing maxterms of canonical POS are the terms present in canonical SOP.
- Maxterms in the expression signify that the output of the circuit assumes a logic 0 for those maxterms/ input combinations.

EXPANSION INTO CANONICAL FORM

► Minterms and their corresponding Maxterms

				<i>Minterms</i>		<i>Maxterms</i>
<i>X</i>	<i>Y</i>	<i>Z</i>		<i>Product Terms</i>		<i>Sum Terms</i>
0	0	0		$m_0 = \bar{X} \cdot \bar{Y} \cdot \bar{Z} = \min(\bar{X}, \bar{Y}, \bar{Z})$		$M_0 = X + Y + Z = \max(X, Y, Z)$
0	0	1		$m_1 = \bar{X} \cdot \bar{Y} \cdot Z = \min(\bar{X}, \bar{Y}, Z)$		$M_1 = X + Y + \bar{Z} = \max(X, Y, \bar{Z})$
0	1	0		$m_2 = \bar{X} \cdot Y \cdot \bar{Z} = \min(\bar{X}, Y, \bar{Z})$		$M_2 = X + \bar{Y} + Z = \max(X, \bar{Y}, Z)$
0	1	1		$m_3 = \bar{X} \cdot Y \cdot Z = \min(\bar{X}, Y, Z)$		$M_3 = X + \bar{Y} + \bar{Z} = \max(X, \bar{Y}, \bar{Z})$
1	0	0		$m_4 = X \cdot \bar{Y} \cdot \bar{Z} = \min(X, \bar{Y}, \bar{Z})$		$M_4 = \bar{X} + Y + Z = \max(\bar{X}, Y, Z)$
1	0	1		$m_5 = X \cdot \bar{Y} \cdot Z = \min(X, \bar{Y}, Z)$		$M_5 = \bar{X} + Y + \bar{Z} = \max(\bar{X}, Y, \bar{Z})$
1	1	0		$m_6 = X \cdot Y \cdot \bar{Z} = \min(X, Y, \bar{Z})$		$M_6 = \bar{X} + \bar{Y} + Z = \max(\bar{X}, \bar{Y}, Z)$
1	1	1		$m_7 = X \cdot Y \cdot Z = \min(X, Y, Z)$		$M_7 = \bar{X} + \bar{Y} + \bar{Z} = \max(\bar{X}, \bar{Y}, \bar{Z})$

KARNAUGH MAPS (K-MAPS)

- ▶ For a 'n' variable input function, number of cells in a K-map will be 2^n
 - ▶ 2-variable, there will be 4 cells
 - ▶ 3-variable, there will be 8 cells and so on
- ▶ Two cells are called Adjacent cells if their minterms differ in only one variable.
- ▶ Grouping of adjacent cells are done in powers of 2 i.e. groups of 2(pair), 4(quad), 8(octet), 16, 32 cells, to write the reduced expression
 - ▶ For expression in SOP, group of 1s are formed
 - ▶ For expression in POS, group of 0s are formed
 - ▶ Generally, SOP is followed, so grouping of 1s are done
- ▶ Cells in a group should be continuous
- ▶ Cells cannot be diagonally combined
- ▶ For minimized expression after grouping, consider the variable that remains constant throughout and ignore the variable that is varying
 - ▶ If variable has constant value 1, then non-complement variable (A)
 - ▶ If variable has constant value 0, then complement variable (A')

2-VARIABLE K-MAP

- ▶ 2-variable K-map will have, $2^2 = 4$ cells. Each cell represents a unique minterm.
- ▶ According to the minterms involved in the output expression/truth table, '1' is placed in that corresponding cell, and '0' otherwise.

	A	B	O/P
m_0	0	0	0
m_1	0	1	1
m_2	1	0	1
m_3	1	1	0

SOP: -

A \ B	$\bar{B}$ 0	B 1
$\bar{A}$ 0	$\bar{A}\bar{B}$	$\bar{A}B$
A 1	$A\bar{B}$	AB

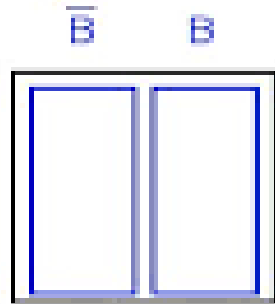
POS: -

A \ B	B 0	$\bar{B}$ 1
A 0	$A+B$	$A+\bar{B}$
$\bar{A}$ 1	$\bar{A}+B$	$\bar{A}+\bar{B}$

A \ B	0	1
0	m_0	m_1
1	m_2	m_3

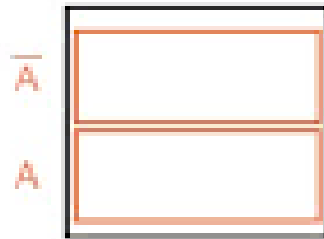
2-VARIABLE K-MAP

- ▶ Singles or Group of 2/4 is formed for reduced expression
- ▶ For minimized expression, consider the variable that remains constant throughout and ignore the variable that varies
 - ▶ If variable has constant value 1, then non-complement variable (A)
 - ▶ If variable has constant value 0, then complement variable (A')



A \ B	0	1
	0	1
0	α	β
1	γ	δ

$$Y = B'$$

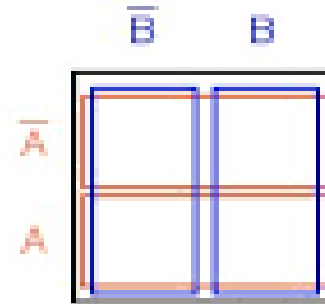


A \ B	0	1
	0	1
0	α	β
1	γ	δ

$$Y = B$$

A \ B	0	1
	0	1
0	α	β
1	γ	δ

$$Y = A'$$



A \ B	0	1
	0	1
0	α	β
1	γ	δ

$$Y = A$$

3-VARIABLE K-MAP

- ▶ 3-variable K-map will have, $2^3 = 8$ cells. Each cell represents a unique minterm.
- ▶ According to the minterms involved in the output expression/truth table, '1' is placed in that corresponding cell, and '0' otherwise.

	A	B	C	O/P
m_0	0	0	0	0
m_1	0	0	1	1
m_2	0	1	0	0
m_3	0	1	1	1
m_4	1	0	0	0
m_5	1	0	1	0
m_6	1	1	0	1
m_7	1	1	1	1

AB \ C	0	1
00	m_0	m_1
01	m_2	m_3
11	m_6	m_7
10	m_4	m_5

A \ BC	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

A \ BC	00	01	11	10
0	$A'B'C'$ ⁰	$A'B'C$ ¹	$A'BC$ ³	$A'BC'$ ²
1	$AB'C'$ ⁴	$AB'C$ ⁵	ABC ⁷	ABC' ⁶

3-VARIABLE K-MAP

- ▶ Singles or Group of 2/4/8 is formed for reduced expression
- ▶ For minimized expression, consider the variable that remains constant throughout and ignore the variable that varies
 - ▶ If variable has constant value 1, then non-complement variable (A)
 - ▶ If variable has constant value 0, then complement variable (A')

A \ BC	BC			
	00	01	11	10
0	0	1	1	1
1	1	1	*	*

$$Y = A + B + C$$

A \ BC	BC			
	00	01	11	10
0	0	0	1	1
1	1	1	*	*

$$Y = A + B$$

A \ BC	BC			
	00	01	11	10
0	0	0	1	0
1	1	1	*	*

$$Y = A + BC$$

A \ BC	BC			
	00	01	11	10
0	0	0	0	0
1	1	1	*	*

$$Y = A$$

A \ BC	BC			
	00	01	11	10
0	0	0	0	0
1	0	1	*	*

$$Y = AC$$

3-VARIABLE K-MAP

- Group of 2 can be formed to obtain the reduced expression

$$Y = A'C' + AC$$

AB \ C	0	1
00	1	0
01	1	0
11	0	1
10	0	1

AB \ C	0	1
00	1	0
01	0	0
11	0	0
10	1	0

$$Y = B'C'$$

- Group of 4 can be formed to obtain the reduced expression

AB \ C	0	1
00	1	1
01	0	0
11	0	0
10	1	1

$$Y = B'$$

AB \ C	0	1
00	1	1
01	1	1
11	0	0
10	0	0

$$Y = A'$$

AB \ C	0	1
00	0	0
01	1	1
11	1	1
10	0	0

$$Y = B$$

AB \ C	0	1
00	0	1
01	0	1
11	0	1
10	0	1

$$Y = C$$

4-VARIABLE K-MAP

- ▶ 4-variable K-map will have, $2^4 = 16$ cells. Each cell represents a unique minterm.
- ▶ According to the minterms involved in the output expression/truth table, '1' is placed in that corresponding cell, and '0' otherwise.

AB \ CD	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

AB \ CD	00	01	11	10
00	$A'B'C'D'$ 0	$A'B'C'D$ 1	$A'B'CD$ 3	$A'B'CD'$ 2
01	$A'BC'D'$ 4	$A'BC'D$ 5	$A'BCD$ 7	$A'BCD'$ 6
11	$ABC'D'$ 12	$ABC'D$ 13	$ABCD$ 15	$ABCD'$ 14
10	$AB'C'D'$ 8	$AB'C'D$ 9	$AB'CD$ 11	$AB'CD'$ 10

4-VARIABLE K-MAP

- ▶ Singles or Group of 2/4/8/16 is formed to obtain the reduced expression
- ▶ For minimized expression, consider the variable that remains constant throughout and ignore the variable that varies
 - ▶ If variable has constant value 1, then non-complement variable (A)
 - ▶ If variable has constant value 0, then complement variable (A')

F

CD	00	01	11	10
AB	1	1	0	1
00	1	1	0	1
01	1	1	0	1
11	1	1	0	0
10	1	1	0	1

$$F = A'D' + B'D' + C'$$

4-VARIABLE K-MAP

	C'D'	C'D	CD	CD'
A'B'	0 1	1 1	3 1	2 0
A'B	4 0	5 1	7 0	6 0
AB	12 1	13 1	15 1	14 0
AB'	8 0	9 1	11 1	10 0

$$Y = A'B'C' + A'B'D + ABC' + C'D + AD$$

AB \ CD	00	01	11	10
00	1	0	0	1
01	0	1	1	0
11	1	1	1	1
10	1	0	0	1

$$Y = AB + BD + B'D$$

AB \ CD	00	01	11	10
00	0	1	1	1
01	1	0	0	1
11	1	0	0	1
10	0	1	1	1

$$Y = B'D + CD' + BD'$$

5-VARIABLE K-MAP

- ▶ 5-variable K-map will have, $2^5 = 32$ cells. Each cell represents a unique minterm.
- ▶ 32 cells are assumed to be divided into two blocks of 16 cells each, such that both blocks overlap/superimpose each other.
 - ▶ One block corresponds to MSB=0 and have minterms from m_0 to m_{15}
 - ▶ One block corresponds to MSB=1 and have minterms from m_{16} to m_{31}
- ▶ Grouping of 2/4/8/16 cells is possible
- ▶ Cells are considered to be adjacent as well as superimposing the cells of other block
- ▶ According to the minterms involved in the output expression/truth table, '1' is placed in that corresponding cell, and '0' otherwise.

5-VARIABLE K-MAP

- ▶ 5-variable K-map will have, $2^5 = 32$ cells.
- ▶ Considering 'A' to be the MSB, one block is defined for 'A=0' and other for 'A=1'

		$\bar{A}$						A			
BC	DE	00	01	11	10	BC	DE	00	01	11	10
00		m ₀	m ₁	m ₃	m ₂	00		m ₁₆	m ₁₇	m ₁₉	m ₁₈
01		m ₄	m ₅	m ₇	m ₆	01		m ₂₀	m ₂₁	m ₂₃	m ₂₂
11		m ₁₂	m ₁₃	m ₁₅	m ₁₄	11		m ₂₈	m ₂₉	m ₃₁	m ₃₀
10		m ₈	m ₉	m ₁₁	m ₁₀	10		m ₂₄	m ₂₅	m ₂₇	m ₂₆

5-VARIABLE K-MAP

		$\bar{A}$						A			
BC	DE	00	01	11	10	BC	DE	00	01	11	10
		00	01	11	10			00	01	11	10
00	00	1	0	0	1	00	00	1	0	0	1
01	01	0	1	1	0	01	01	0	1	1	0
11	11	0	1	1	0	11	11	0	1	1	0
10	10	1	0	0	1	10	10	1	0	0	1

$$F = CE + C'E'$$

Groups of 8

$$F = D'$$

		$\bar{A}$						A			
BC	DE	00	01	11	10	BC	DE	00	01	11	10
		00	01	11	10			00	01	11	10
00	00	1	1	0	0	00	00	1	1	0	0
01	01	1	1	0	0	01	01	1	1	0	0
11	11	1	1	0	0	11	11	1	1	0	0
10	10	1	1	0	0	10	10	1	1	0	0

Group of 16

DON'T CARE COMBINATION

- ▶ The values or the input combinations for which the output function is unspecified are called DON'T CARE COMBINATIONS
- ▶ They are denoted by 'd' or (*) or 'X' in the expression and K-maps
- ▶ SOP: During grouping, each don't care is treated as '1' if it is helpful in reducing the expression, otherwise it is considered '0'.
- ▶ POS: During grouping, each don't care is treated as '0' if it is helpful in reducing the expression, otherwise it is considered '1'.
- ▶ E.g.: $F1 = \Sigma(0,1,2,3,4,5) + d(6,7)$
 $F2 = \Sigma(2,3,4,5) + d(6,7)$

		BC			
A		00	01	11	10
	0	0	1	1	1
	1	1	1	*	*

$$F1 = A + B + C$$

		BC			
A		00	01	11	10
	0	0	0	1	1
	1	1	1	*	*

$$F2 = A + B$$

Problems part-1

1. Expand $[A(A'+B)(A'+B+C')]$ to maxterms and minterms
2. Find the reduced expression using K-maps:
 1. $F = A'B' + A'B + AB'$
 2. $F = A'B' + AB + A'B$
 3. $F = (A+B)(A+B')(A'+B')$
 4. $F = A'B'C' + A'BC' + ABC' + AB'C$
 5. $F = (A+B+C)(A'+B+C')(A'+B'+C')(A+B'+C')(A'+B'+C)$
 6. $F = AB' + AC' + ABC + AB'$
(First convert to canonical SOP & then reduce)
 7. $F(A,B,C) = \sum m(0,4,5,7)$
 8. $F(A,B,C) = \sum m(0,1,3,5,7)$
 9. $F(A,B,C,D) = \sum m(0,1,2,3,5,7,8,9,10,12,13)$
 10. $F(A,B,C,D,E) = \sum m(0,2,3,10,11,12,13,16,17,18,19,20,21,26,27)$
 11. $F(A,BC,D) = \sum m(0,2,6,10,11,12,13) + d(3,4,5,14,15)$

Problems part-2

Obtain the minimal expression using K-map

1. $F(A,B,C,D) = \sum m(0,1,3,4,5,7,10,13,14,15)$
2. $F(A,B,C,D) = \sum m(0,2,3,6,7,8,10,11,12,15)$
3. $F(A,B,C,D) = \sum m(0,1,3,4,5,6,7,13,15)$
4. $F(A,B,C,D) = \prod M(6,7,8,9). d(10,11,12,13,14,15)$
5. $F(A,B,C,D) = \prod M(1,5,6,7,11,12,13,15)$
6. $F(A,B,C,D) = \sum m(1,5,6,12,13,14) + d(2,4)$

Problems

1. Expand $[A(A'+B)(A'+B+C')]$ to maxterms and minterms =
2. Find the reduced expression using K-maps:
 1. $F = A'B' + A'B + AB' = A' + B'$
 2. $F = A'B' + AB + A'B = A' + B$
 3. $F = (A+B)(A+B')(A'+B') = AB'$
 4. $F = A'B'C' + A'BC' + ABC' + AB'C = A'C' + BC' + AB'C$
 5. $F = (A+B+C)(A'+B+C')(A'+B'+C')(A+B'+C')(A'+B'+C)$
 6. $F = AB' + AC' + ABC + AB'$ (First convert to canonical SOP & then reduce)
 7. $F(A,B,C) = \sum m(0,4,5,7) = AC + B'C'$
 8. $F(A,B,C) = \sum m(0,1,3,5,7) = C + A'B'$
 9. $F(A,B,C,D) = \sum m(0,1,2,3,5,7,8,9,10,12,13) = B'D' + AC' + A'D$
 10. $F(A,B,C,D,E) = \sum m(0,2,3,10,11,12,13,16,17,18,19,20,21,26,27)$
 11. $F(A,B,C,D) = \sum m(0,2,6,10,11,12,13) + d(3,4,5,14,15)$